
IAL STATISTICS 1 · A* EXAMINATION GUIDE

Pearson Edexcel International Advanced Level · Paper WST01/01

Built from 18 question papers and their mark schemes, October 2020 – Summer 2026

How to use this guide. Each topic gives the facts you must own, the method examiners reward, the exact wording that earns (and loses) the one-mark “interpret” and “comment” marks, and real past-paper questions solved the way the scheme awards them. In S1 the calculations are rarely the problem; the marks leak through wording, rounding and the use of tables. Read Section 2 first.

1. THE PAPER AT A GLANCE

Duration and total	1 hour 30 minutes, 75 marks, 6 to 8 questions (≈ 1.2 minutes per mark)
Tables	Normal distribution table and percentage-points table in the formula booklet
Accuracy	3 significant figures unless told otherwise; probabilities from the normal table to 4 d.p.
Shape of the paper	Every paper has one question on each of: probability, a discrete random variable, the normal distribution, correlation and regression, and data presentation

Topic frequency across the 18 papers analysed

TOPIC	PAPERS	MARKS	USUAL FORM
Discrete random variables	18/18	6–15	Find unknown probabilities; $E(X)$, $\text{Var}(X)$, $\text{Var}(aX + b)$; probabilities of inequalities in X
Normal distribution	18/18	7–15	Standardise; inverse with percentage points; find μ and σ ; conditional probability; “exactly 2 of 4”
Correlation and regression	18/18	8–17	S_{xy} etc., r , line of y on x , interpretations, reliability, effect of coding
Probability: Venn diagrams	15/18	6–12	Unknown regions from independence or conditional statements
Probability: tree diagrams	13/18	6–11	With and without replacement; reverse conditional “given that”
Outliers	14/18	2–4	$Q_1 - 1.5\text{IQR}$, $Q_3 + 1.5\text{IQR}$; sometimes on a normal model
Histograms	11/18	3–10	Bar width and height from a given bar; frequency from area; interpolation
Box plots	10/18	3–5	Draw with outliers; compare two distributions
Stem and leaf (incl. back-to-back)	9/18	3–6	Median and quartiles; missing values from given statistics
Cumulative distribution function	9/18	2–6	Convert $F(x)$ to $P(X = x)$ and back
Skewness	8/18	1–2	Quartile, mean–median or formula measure; describe and interpret
Coding	7/18	3–6	Mean, variance, r and regression line under $y = \frac{x-a}{b}$
Discrete uniform	7/18	1–6	Name it; use symmetry for $E(X)$

2. HOW THE MARK SCHEME AWARDS MARKS

M1	Method: e.g. “standardising 10 with 18 and 5.4 (allow $\pm$)”, “correct method for r using their S_{yy} ”.
B1	An independent value or statement: a percentage-point z value, an interpretation, a reason.
A1, awrt	Answer, usually “awrt” (answers which round to) a 3 s.f. value. cs0 : correct solution only.
*	Given answer: needs a correct numerical expression with no incorrect working.
ft	Follow-through: a correct method on your wrong earlier value still scores.

RULINGS THAT RECUR ACROSS THE 18 SCHEMES

- **Percentage-point z values carry a B1 and must be to 4 d.p.** (0.8416, 1.2816, 1.6449, ...). In October 2020, the answer $d = 3.04$ without $z = 0.8416$ seen scored only M1 B0 A1.
- **Show the standardisation.** Recent papers print “use standardisation” or “solutions relying entirely on calculator technology are not acceptable”; a bare calculator answer then scores nothing.
- **Interpretations must be in context.** “Strong positive correlation between population and GDP” on its own was B0; “extrapolation so unreliable” on its own was B0.
- Using $\sum y^2$ in place of S_{yy} in the formula for r is M0.
- Regression coefficients: decimals only (“do not allow fractions”), the question’s own letters (g and t , not y and x), and the given precision (46.0, not 46).
- Venn diagrams: every region filled in, including the region outside all the events; a box for the sample space.
- Independence: a numerical comparison *and* a stated conclusion are both needed.
- A “positive skew” label where the question wanted “positive correlation” (or vice versa) is B0.

3. REPRESENTING AND SUMMARISING DATA

FACTS TO OWN

Quartiles from raw data (stem and leaf, n values in order): median is the $\frac{n+1}{2}$ th value. For Q_1 , find $\frac{n}{4}$: if not a whole number, round *up* and take that value; if a whole number, average that value and the next. Q_3 likewise from $\frac{3n}{4}$.

Grouped data (tables, histograms): use $\frac{n}{2}$, $\frac{n}{4}$, $\frac{3n}{4}$ directly with **linear interpolation**: $Q = L + \frac{\text{position} - \text{cum. freq. before}}{\text{class freq.}} \times \text{class width}$.

Outliers: below $Q_1 - k(Q_3 - Q_1)$ or above $Q_3 + k(Q_3 - Q_1)$, with k given (usually 1.5). Sometimes “mean ± 2 or 3 s.d.”.

Box plot: box from Q_1 to Q_3 with the median; whiskers to the last values that are not outliers (whiskers to the outlier limits are also accepted); each outlier marked $\times$.

Histogram: area $\propto$ frequency; frequency density = frequency $\div$ class width. Used because the data are *continuous*.

Skewness: positive if $Q_3 - Q_2 > Q_2 - Q_1$, or mean $>$ median, or a formula value > 0 . Positive skew means the tail is on the high side: values are more spread out above the median.

Method for histogram bar questions

1. Use the given bar to find the scale: width in cm per unit of the variable, and cm^2 per unit of frequency.
2. For the new bar: width from the width scale; height = (its frequency $\times$ cm^2 per unit) $\div$ its width in cm.
3. Check class boundaries: $20 < l \leq 25$ has width 5; “11–20 hours to the nearest hour” runs from 10.5 to 20.5.

Worked example — Summer 2022, Q3(a)(b) (5 marks). 88 logs. The bar for $27 < l \leq 30$ (frequency 15) is 1.5 cm wide and 4 cm high. Find the bar for $20 < l \leq 25$ (frequency 35), then the median. Cumulative frequency to 20 is 19.

WORKING	MARK
Width: 3 units $\rightarrow$ 1.5 cm, so 5 units $\rightarrow$ 2.5 cm	B1
Area: $1.5 \times 4 = 6 \text{ cm}^2$ for 15 logs, so 35 logs $\rightarrow 14 \text{ cm}^2$; height = $14 \div 2.5 = 5.6 \text{ cm}$	M1 A1
Median at $\frac{88}{2} = 44\text{th}$: $20 + \frac{44-19}{35} \times 5$	M1
= 23.6 cm (using $n + 1$, i.e. 44.5th, was also allowed)	A1

Worked example — October 2023, Q2 (13 marks). 33 kangaroo weights (kg) in a stem and leaf diagram. Find the quartiles, show there are two outliers, draw the box plot and compare it with February’s.

WORKING	MARK
Median: 17th value = 57; $Q_1: \frac{33}{4} = 8.25 \rightarrow 9\text{th} = 45$; $Q_3: 24.75 \rightarrow 25\text{th} = 63$	B1 B1 B1
Limits: $63 + 1.5(18) = 90$; $45 - 1.5(18) = 18$	M1 A1ft
16 and 94 lie outside, so they are the outliers	A1*
Box with two whiskers; Q_1, Q_2, Q_3 correct; whiskers to 23 and 86 (or 18 and 90); $\times$ at 16 and 94	M1 A1ft A1ft A1
Compare the medians (must say “median”), compare the IQR or range, then interpret one in context: “on average the kangaroos weighed less in February”	B1ft B1ft B1ft

EXAMINER TRAPS

- “Comment on differences” needs a named statistic with its value; comments on skew or outliers were ignored.
- “Describe and interpret the skew”: positive skew *plus* a context statement (“weights are more varied above the median”).
- “Explain why the answers are estimates”: the data are grouped, so the exact values are not known (mid-points are used).
- A normal model is unsuitable for clearly skewed data; a symmetric box plot supports it.

4. MEAN, VARIANCE AND CODING

FACTS TO OWN

$$\bar{x} = \frac{\sum x}{n} \quad \sigma^2 = \frac{\sum x^2}{n} - \bar{x}^2 = \frac{S_{xx}}{n} \quad \text{grouped: } \bar{x} = \frac{\sum fx}{\sum f}, \sigma^2 = \frac{\sum fx^2}{\sum f} - \bar{x}^2 \text{ (mid-points).}$$

Coding $y = \frac{x-a}{b}$: $\bar{x} = a + b\bar{y}$, $\sigma_x = |b|\sigma_y$, $\sigma_x^2 = b^2\sigma_y^2$. Adding a constant changes the mean, not the spread.

Combining groups: add the totals: $\sum x = n_1\bar{x}_1 + n_2\bar{x}_2$, and $\sum x^2$ by adding (or by $\sum x^2 = n(\sigma^2 + \bar{x}^2)$ for each group).

Adding a value equal to the mean: the mean is unchanged and the standard deviation *decreases* ($\sum (x - \bar{x})^2$ is unchanged but n increases).

Worked example — Summer 2022, Q3(e) (6 marks). Log weights coded $y = 0.5w - 255$; $n = 88$, $\sum y = 924$, $\sum y^2 = 12\,862$. Find the mean and variance of W .

WORKING	MARK
$\bar{y} = 924/88 = 10.5$	M1
$w = 2(y + 255)$, so $\bar{w} = 2(10.5 + 255) = 531$ g	M1 A1
$\sigma_y^2 = \frac{12\,862}{88} - 10.5^2 = 35.909\dots$	M1
$\sigma_w^2 = 2^2 \times 35.909\dots = 144$ g ²	M1 A1

Worked example — Summer 2025, Q2 (4 marks). Class A: $n = 24$, $\bar{x} = 47$, $\sum x^2 = 66\,876$. Class B: $n = 30$, mean b , $\sum x^2 = 73\,826$. The combined mean is 45. Find b and the combined s.d.

WORKING	MARK
$24(47) + 30b = 54(45) \Rightarrow 1128 + 30b = 2430 \Rightarrow b = 43.4$	M1 A1*
$\sigma^2 = \frac{66\,876 + 73\,826}{54} - 45^2 = 580.59 \dots, \sigma = 24.1$	M1 A1

Part (c) then asked for effects of adding 2 to every class B mark: variance of B unchanged (*reason*: spread is unchanged by adding a constant; “not affected by coding” alone was B0); combined mean increases (the total increases by 60).

5. CORRELATION AND REGRESSION

FACTS TO OWN $S_{xx} = \sum x^2 - \frac{(\sum x)^2}{n} \quad S_{xy} = \sum xy - \frac{\sum x \sum y}{n} \quad r = \frac{S_{xy}}{\sqrt{S_{xx}S_{yy}}} \quad b = \frac{S_{xy}}{S_{xx}}, \quad a = \bar{y} - b\bar{x}.$ The regression line of y on x passes through $(\bar{x}, \bar{y})$; two regression lines meet at $(\bar{x}, \bar{y})$. Use y on x only to predict y from x. x is the explanatory (independent, controlled) variable; y is the response. Coding: linear coding does not change r (multiplying one variable by a negative flips its sign). For the regression line, substitute the codes and rearrange. Changes to data: doubling w leaves r and the intercept unchanged and halves the gradient; adding the point $(\bar{x}, \bar{y})$ leaves S_{xx}, S_{yy}, S_{xy} and so r unchanged.
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QUESTION ASKS	WORDING THAT SCORED THE MARK
Interpret r	“As height increases, weight tends to increase” (variables in context; no need to quote r). “Strong positive correlation” alone: B0
Consistent with a linear model?	“Yes, $r = 0.953$ is close to 1, so the points lie close to a straight line”
Interpret the gradient	“For each extra 1 cm of height, weight increases by 3.27 kg (on average)”
Interpret the intercept	The value of y when $x = 0$, in context; say if $x = 0$ is meaningless or outside the data
Reliability of an estimate	“153 cm is within the range of heights in the data (138–157), so it is interpolation and reliable”; “170 cm is outside the range, extrapolation”. Must refer to the x -values
Why use statistical models	They simplify the real world to aid understanding; they can be refined; they show relationships. (Cost, time or “easy” alone: B0)

Worked example — October 2025, Q1 (12 marks). 8 horses: $\sum h = 1160$, $\sum w = 4160$, $\sum hw = 604\,135$, $\sum w^2 = 2\,166\,600$, $S_{hh} = 286$.

WORKING	MARK
$S_{hw} = 604\,135 - \frac{1160 \times 4160}{8} = 935$; $S_{ww} = 2\,166\,600 - \frac{4160^2}{8} = 3400$	M1 A1 A1
$r = \frac{935}{\sqrt{286 \times 3400}} = 0.948$	M1 A1
$b = \frac{935}{286} = 3.269 \dots$; $a = 520 - 3.269 \dots (145) = 45.96 \dots$	M1 M1
$w = 46.0 + 3.27h$ (“46 + 3.27h” was not accepted)	A1*
For each extra 1 cm of height, weight increases by 3.27 kg	B1
$h = 153$: $w = 546$ kg; $h = 170$: $w = 602$ kg	B1 B1
153 is more reliable: it lies within the data’s heights (interpolation)	B1

EXAMINER TRAPS

- Coded regression: if $w = 11.5d$ and $w = 30.9 - 3.46t$, then $d = 2.69 - 0.301t$; do not forget to divide every term.
- “Range of x for which the model predicts $f > p$ ”: solve the inequality, then intersect with the range of the data.
- Keep S_{xy} etc. unrounded in the calculator; rounding b before finding a spoils the given answer.

6. PROBABILITY

FACTS TO OWN

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) \quad P(A | B) = \frac{P(A \cap B)}{P(B)} \quad P(A') = 1 - P(A).$$

Independent: $P(A \cap B) = P(A)P(B)$, equivalently $P(A | B) = P(A)$. If A, B are independent then so are A' and B , etc.

Mutually exclusive: $P(A \cap B) = 0$; drawn as non-overlapping circles. $P(C) + P(D) > 1$ proves C, D cannot be mutually exclusive.

Tree diagrams: multiply along branches, add between branches. Reverse (Bayes-type) conditional = wanted path $\div$ total of all paths giving the condition.

Method

1. Venn diagram with unknowns: write every region in terms of the letters, use “total = 1”, then one equation from each independence or conditional statement.
2. Without replacement: denominators fall by one each draw; algebra in n gives a quadratic, so reject non-integer or negative roots and say why.
3. “Given that” always means a division: identify the denominator before calculating.

Worked example — Summer 2025, Q4 (4 marks). $P(C \cup D) = 0.59$, $P(D) = 0.45$, $P(C | D) = 0.2$. Find $P(C \cap D)$ and $P(C)$.

WORKING	MARK
$P(C \cap D) = P(C D)P(D) = 0.2 \times 0.45 = 0.09$	M1 A1
$P(C) = 0.59 - 0.45 + 0.09 = 0.23$	M1 A1

Worked example — Summer 2023, Q4 (9 marks). Counters labelled A (30%), B (45%), C (25%); 2%, 4%, 6% respectively are red.

WORKING	MARK
Tree completed: C branch 0.25; red / not red pairs 0.02/0.98, 0.04/0.96, 0.06/0.94	B1 B1
$P(A \cap R') = 0.3 \times 0.98 = 0.294$	M1 A1
$P(R) = 0.3(0.02) + 0.45(0.04) + 0.25(0.06) = 0.039$	M1 A1
$P(C R) = \frac{0.015}{0.039} = \frac{5}{13} = 0.385$	M1 A1ft A1

EXAMINER TRAPS

- Independence questions need the comparison *and* the words “so (not) independent”.
- Venn diagram from survey counts: start from the centre (all three) and work outwards; remember the people outside every circle.
- When a game “ends” on a branch (e.g. a yellow counter first), that branch has no second stage.

7. DISCRETE RANDOM VARIABLES

FACTS TO OWN

$$\sum P(X = x) = 1 \quad E(X) = \sum xP(X = x) \quad E(X^2) = \sum x^2P(X = x) \quad \text{Var}(X) = E(X^2) - [E(X)]^2.$$

$$E(aX + b) = aE(X) + b \quad \text{Var}(aX + b) = a^2 \text{Var}(X) \quad (\text{so } \text{Var}(5 - 2X) = 4 \text{Var}(X)).$$

c.d.f.: $F(x) = P(X \leq x)$; $P(X = x) = F(x) - F(x - 1)$; the last value of F is 1.

Discrete uniform on $1, \dots, n$: $E(X) = \frac{n+1}{2}$, $\text{Var}(X) = \frac{n^2-1}{12}$. Any symmetric distribution has $E(X)$ at the centre (“write down” questions).

Method

1. Inequalities in X : solve for X first, then add the probabilities ($3X - 3 < X + 2 \Rightarrow X < 2.5$).
2. Two independent variables (two spinners, two players): list the pairs that satisfy the condition and add the products.
3. Unknown probabilities: use $\sum P = 1$ and the given $E(X)$ (and $\text{Var}(X)$) as simultaneous equations.
4. New variable such as $Y = \frac{1}{X}$ or $D = |R - B|$: tabulate its values first, combining equal values.

Worked example — Summer 2025, Q1 (9 marks). $P(X = 1, 2, 3, 4) = 0.3, 0.2, 0.35, 0.15$.

WORKING	MARK
$F(3.5) = P(X \leq 3) = 0.85$	B1
$3X - 3 < X + 2 \Rightarrow X < 2.5: P(X \leq 2) = 0.5$	M1 A1
$E(X) = 0.3 + 0.4 + 1.05 + 0.6 = 2.35$	M1 A1
$E(X^2) = 0.3 + 0.8 + 3.15 + 2.4 = 6.65; \text{Var}(X) = 6.65 - 2.35^2 = 1.1275$	M1 M1 A1*
$\text{Var}(5 - 2X) = 4(1.1275) = 4.51$	B1

EXAMINER TRAPS

- $\text{Var}(aX + b)$: the b disappears and a is squared, including a negative a .
- “Show that $\text{Var}(X) = \dots$ ”: write the $E(X^2)$ sum in full before subtracting.
- From a c.d.f. with a constant k : use $F(\text{largest}) = 1$ first.

8. THE NORMAL DISTRIBUTION

FACTS TO OWN

$X \sim N(\mu, \sigma^2)$: the second parameter is the *variance* ($N(443, 9)$ has $\sigma = 3$). $Z = \frac{X - \mu}{\sigma}$.

Symmetry: $P(Z < -a) = 1 - \Phi(a)$; $P(\mu - k < X < \mu + k) = p \Rightarrow \Phi\left(\frac{k}{\sigma}\right) = \frac{1+p}{2}$.

Percentage points to 4 d.p.:

Upper tail p	0.40	0.30	0.25	0.20	0.15	0.10	0.05	0.025
z	0.2533	0.5244	0.6745	0.8416	1.0364	1.2816	1.6449	1.9600

Method

1. Sketch the curve and shade the region; decide which side of the mean each value lies.
2. Inverse problems: write $\frac{x - \mu}{\sigma} = \pm z$ with the sign matching the side of the mean.
3. Two unknowns μ, σ : form two such equations and solve simultaneously.
4. Conditional: $P(A | B) = \frac{P(A \cap B)}{P(B)}$ with $A \cap B$ found from the sketch. A “heavy letter” or “a fruit not moved” is a condition.
5. Several independent items: “exactly 2 of 4” = $\binom{4}{2}p^2(1-p)^2$; “at least one” = $1 - (1-p)^n$.
6. “Correct to the nearest 50 kg is 700” means $675 \leq S < 725$.

Worked example — October 2023, Q5(e) (5 marks). $Y \sim N(\mu, \sigma^2)$; 5% weigh less than 152 g and 40% more than 180 g.

WORKING	MARK
$\frac{152 - \mu}{\sigma} = -1.6449$	M1 A1
$\frac{180 - \mu}{\sigma} = 0.2533$	A1
Subtract: $28 = 1.8982\sigma$	dM1
$\sigma = 14.8, \mu = 176$	A1

Worked example — January 2026, Q7(c) (4 marks). $P(Y < \mu + 5) = 0.7967$. Find $P(Y < \mu - 5 \mid Y < \mu)$ and show $\sigma = 6$.

WORKING	MARK
By symmetry $P(Y < \mu - 5) = 1 - 0.7967 = 0.2033$; divide by $P(Y < \mu) = 0.5$: 0.407	M1 A1
$\Phi(z) = 0.7967 \Rightarrow \frac{5}{\sigma} = 0.83 \Rightarrow \sigma = 6.02\dots$, so $\sigma = 6$	M1 A1*

EXAMINER TRAPS

- Using z to 2 d.p. when the percentage-points table gives 4 d.p. loses the B1.
- Using the variance as σ , e.g. $N(4.5, 0.4^2)$ has $\sigma = 0.4$, but $N(443, 9)$ has $\sigma = 3$.
- Wrong sign of z for a value below the mean.
- “Comment on using a normal model”: it needs a symmetric distribution; quote the skew or the mean–median comparison as evidence.

9. RECENT TRENDS, 2024 – 2026

The five core question types are fixed, but recent papers add twists that reward understanding over routine.

PAPER	NEW TWIST
Summer 2026	$N(443, 9)$ given with variance; a data point at $(\bar{x}, \bar{y})$ added and the effect on r asked without calculation; $\text{Var}(4 - 2X)$ written down from $\text{Var}(2X + 4)$
January 2026	Effect of three corrected marks on a box plot; “which is the response variable?”; $\text{Var}(3 - aX)$ found after a comes from $E(aX + 5) = 6$
October 2025	$P(B \mid C')$ from a Venn diagram; a tree diagram rebuilt from reverse conditional information; conditional normal between two bounds
Summer 2025	“Give one other reason why statistical models are used”; coded regression $w = 11.5d$; $P(X > k) = P(Y < 2k)$ across two normal models
January 2025	Normal value “correct to the nearest 50 kg”; effect of doubling a variable on r , gradient and intercept; does a box plot support a normal model?
October 2024	“Justify the use of a histogram”; normal model judged against skewed data; mean, mode and s.d. skewness measure from a back-to-back stem and leaf
Summer 2024	Missing values constrained by an unchanged box plot; $P(T > \mu - 5 \mid T > \mu - 10)$

10. THE A* CHECKLIST

BEFORE THE EXAM

- Work every one of the 18 papers timed, then mark each “interpret” and “comment” answer against the scheme’s exact wording.
- Learn the percentage-point z values in Section 8 and the interpretation templates in Section 5.
- Practise reading the normal table both ways quickly; know where the calculator’s distribution functions are, but always write the standardisation.

IN THE EXAM ROOM

1. **Write the formula, then the substitution, then the answer:** method marks sit on the substitution line.
2. Keep full calculator values; round only the final answer (3 s.f., or as asked).
3. Every comment in context: name both variables and the units.
4. For any normal question: sketch, standardise in writing, use 4 d.p. z values.
5. For every Venn diagram: fill every region, including outside, and check the total is 1 (or n).
6. "Show that": a complete numerical expression, no incorrect working, final line exactly as printed.
7. Budget about 1.2 minutes per mark; a long probability question stuck at an equation should be left and revisited.